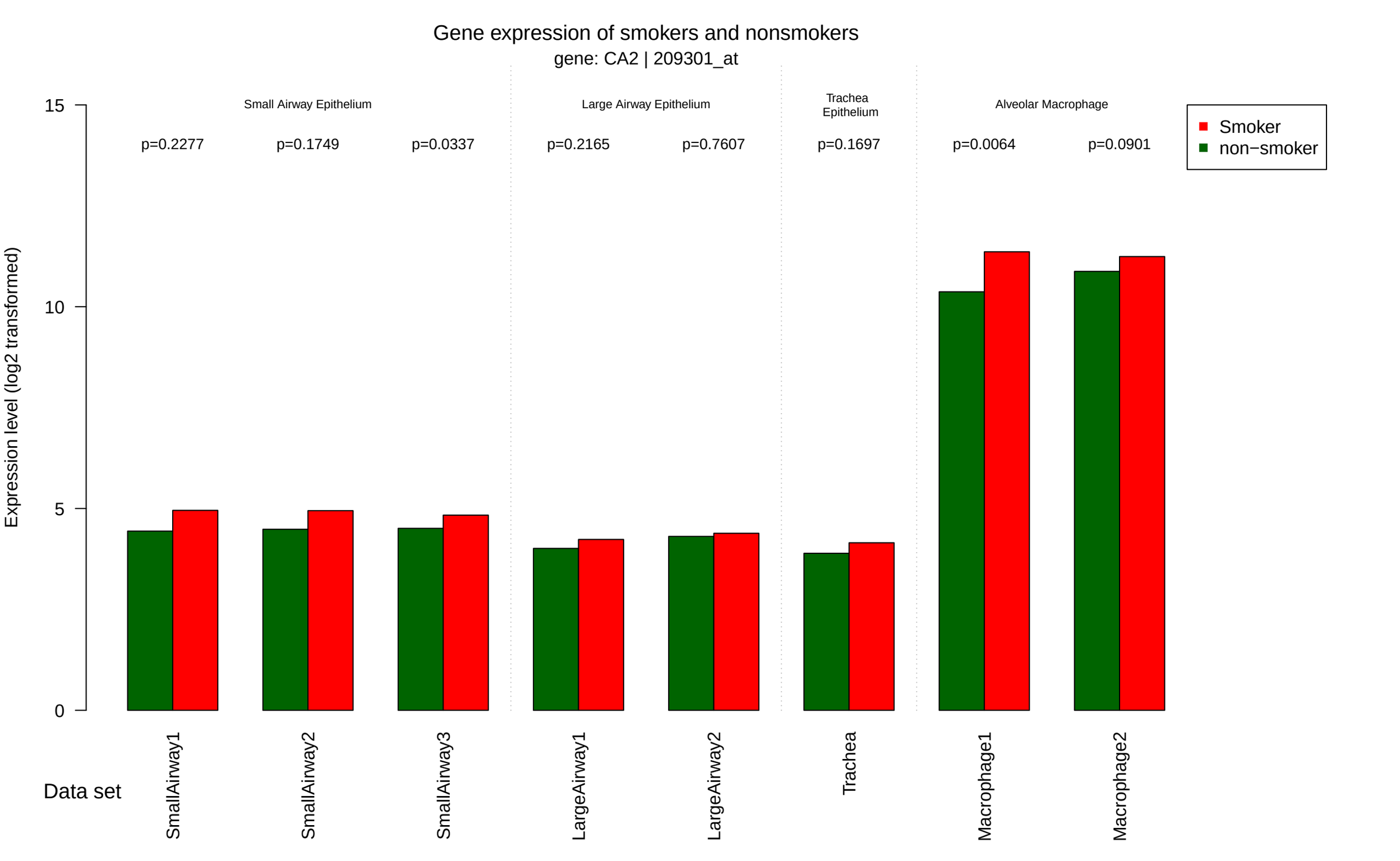
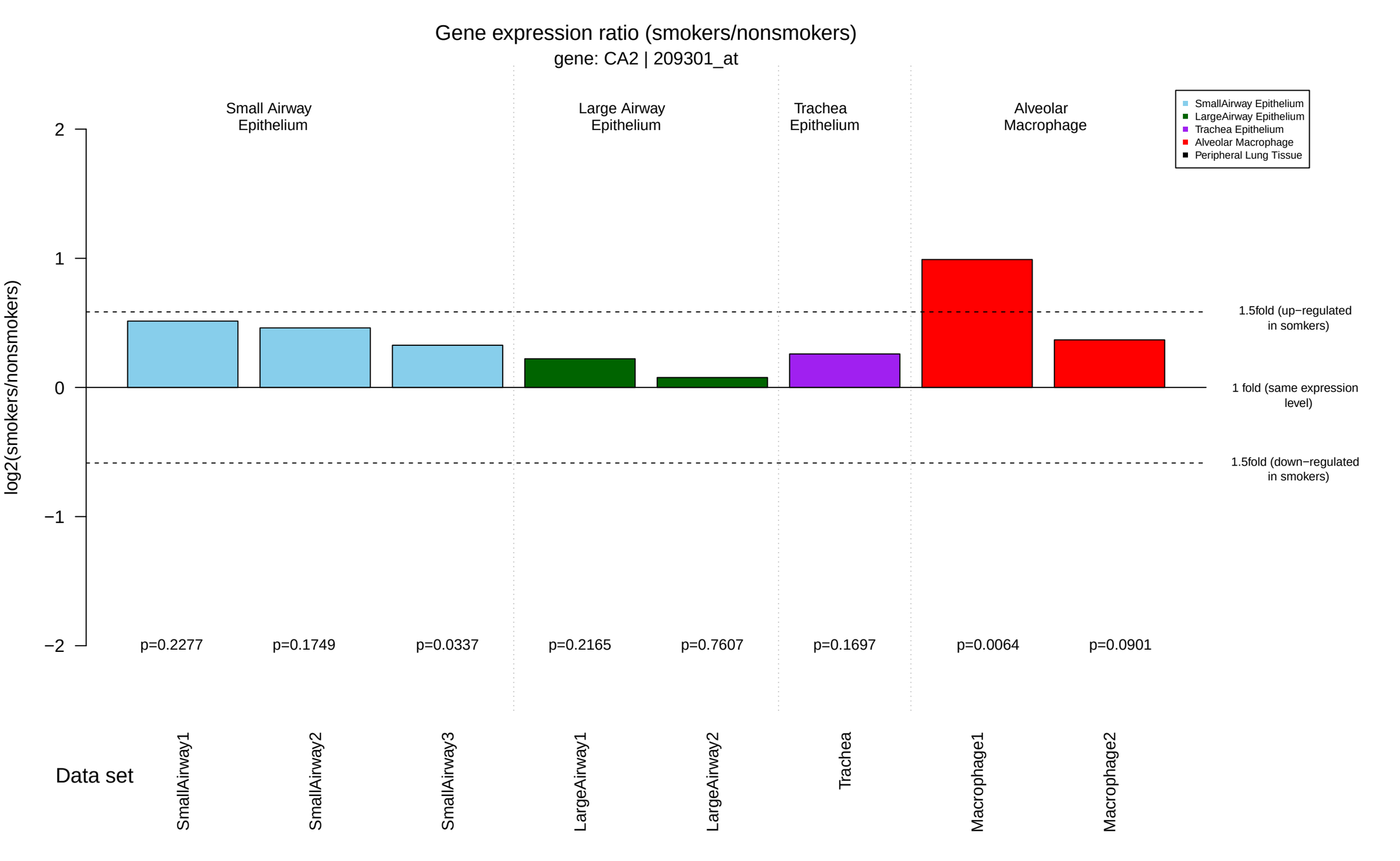
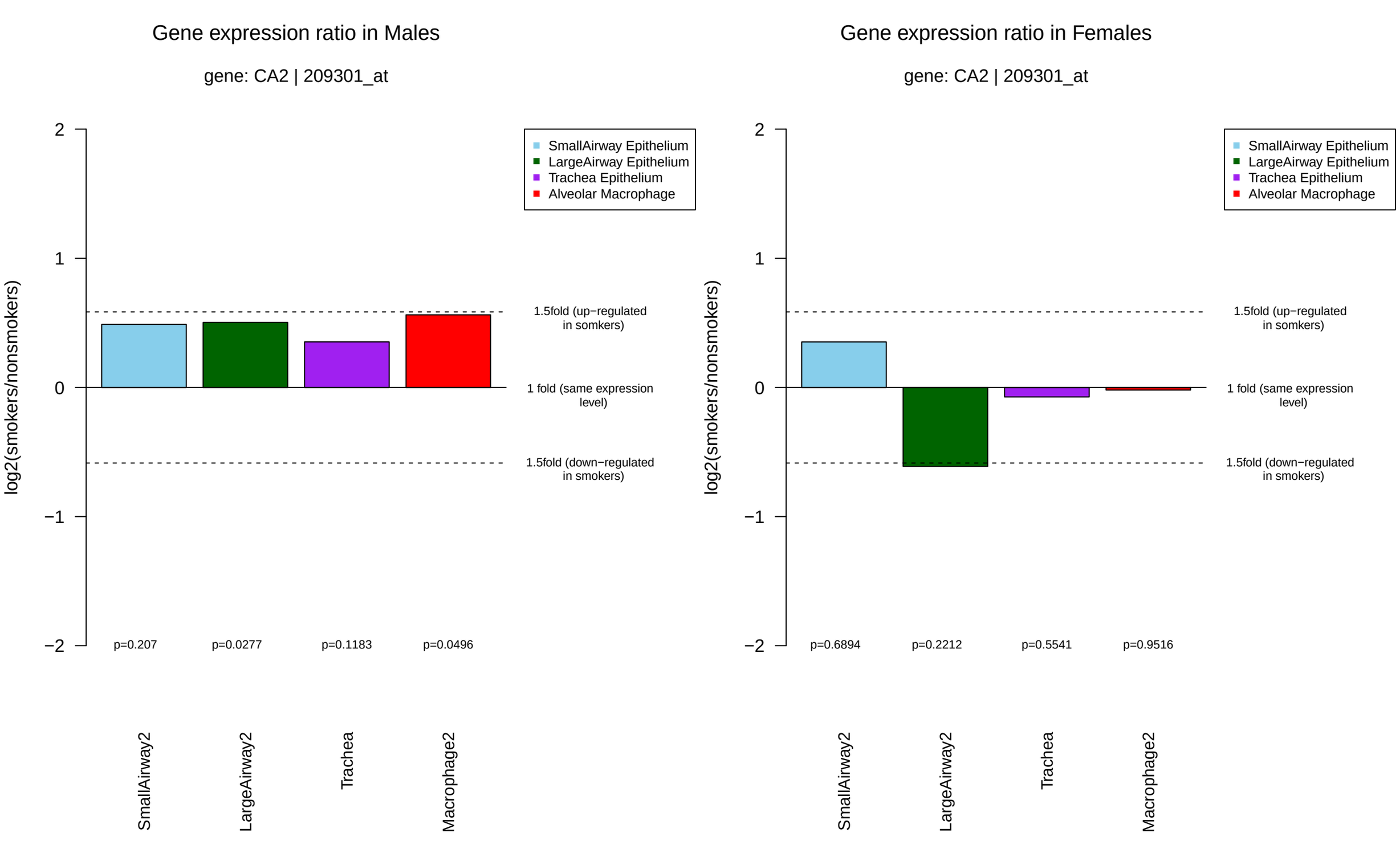
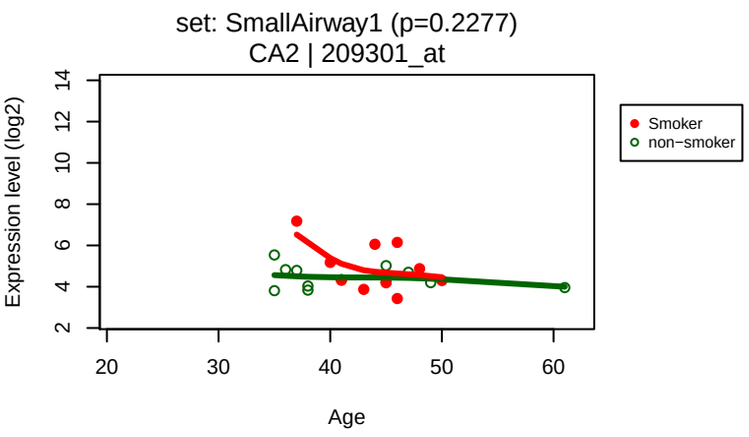
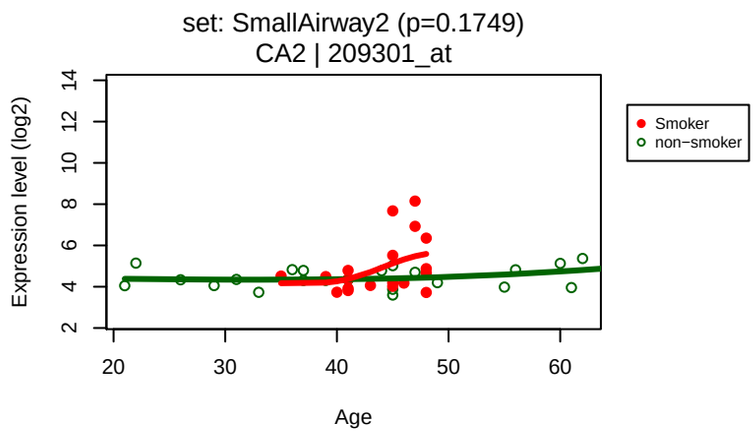
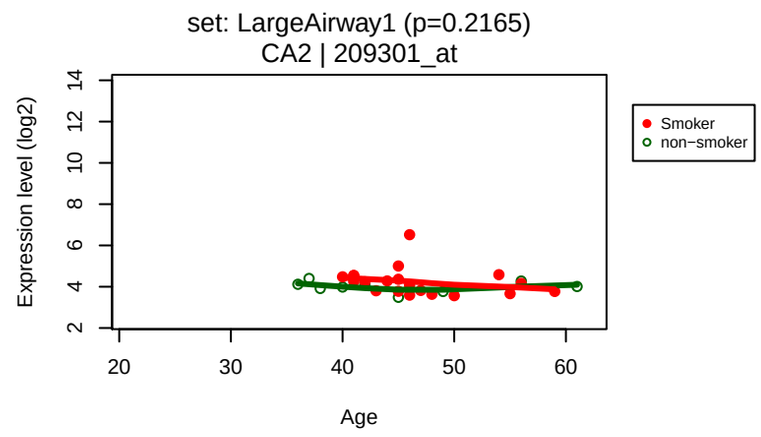

Gene expression of smokers and nonsmokers
gene: CA2 | 209301_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2277
p=0.1749
p=0.0337
p=0.2165
p=0.7607
p=0.1697
p=0.0064
p=0.0901
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

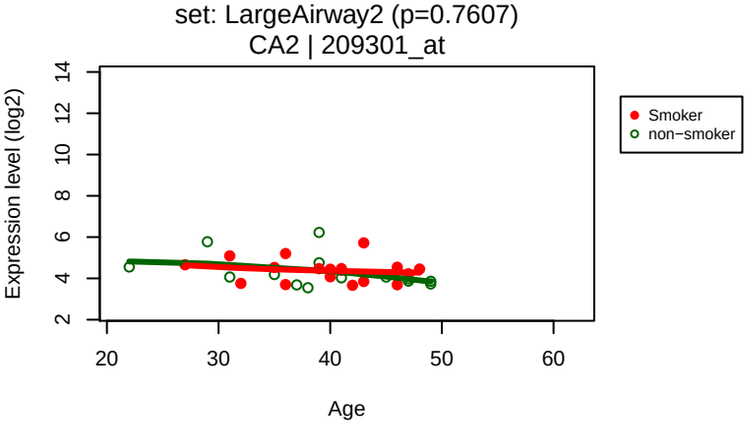

Gene expression ratio (smokers/nonsmokers)
gene: CA2 | 209301_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2277
p=0.1749
p=0.0337
p=0.2165
p=0.7607
p=0.1697
p=0.0064
p=0.0901
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

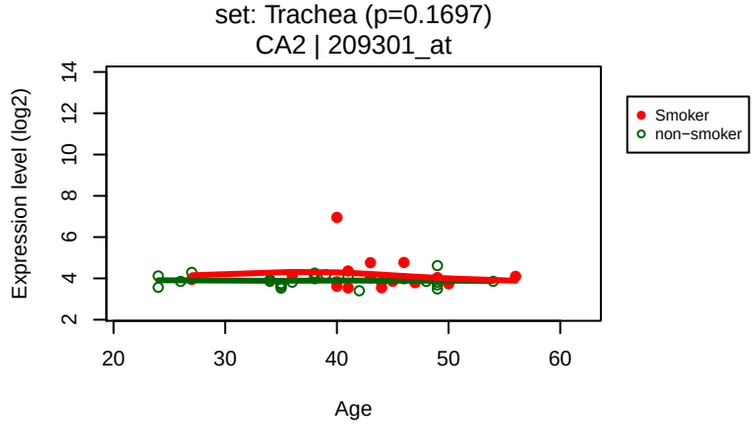

Gene expression ratio in Males
gene: CA2 | 209301_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.207
p=0.0277
p=0.1183
p=0.0496
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA2 | 209301_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6894
p=0.2212
p=0.5541
p=0.9516
SmallAirway2
LargeAirway2
Trachea
Macrophage2

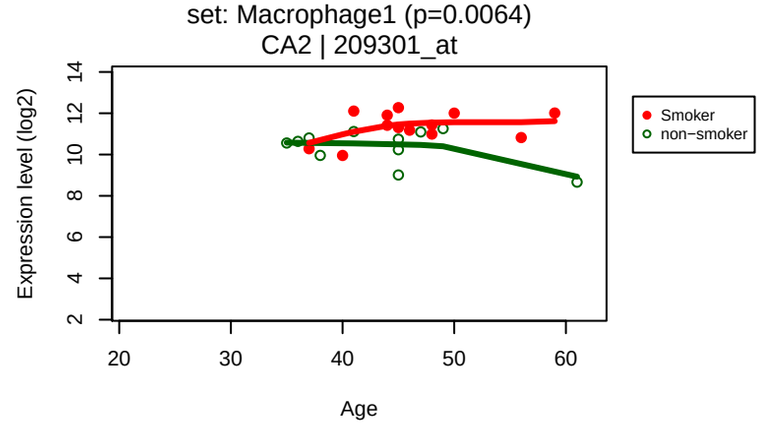

set: SmallAirway1 (p=0.2277)
CA2 | 209301_at
Expression level (log2)
Age
Smoker
non−smoker

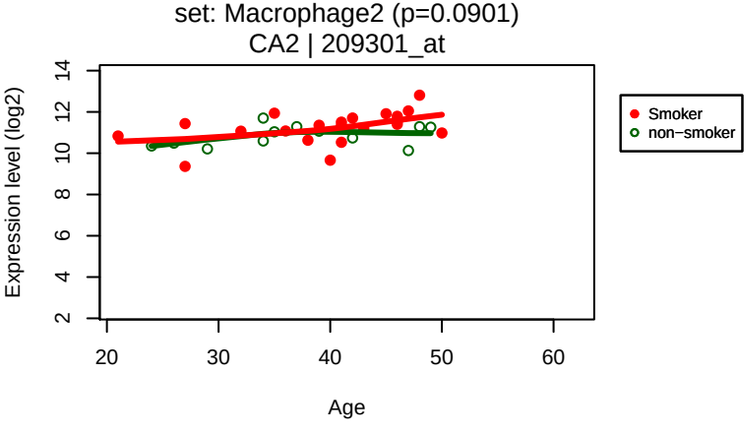

set: SmallAirway2 (p=0.1749)
CA2 | 209301_at
Expression level (log2)
Age
Smoker
non−smoker

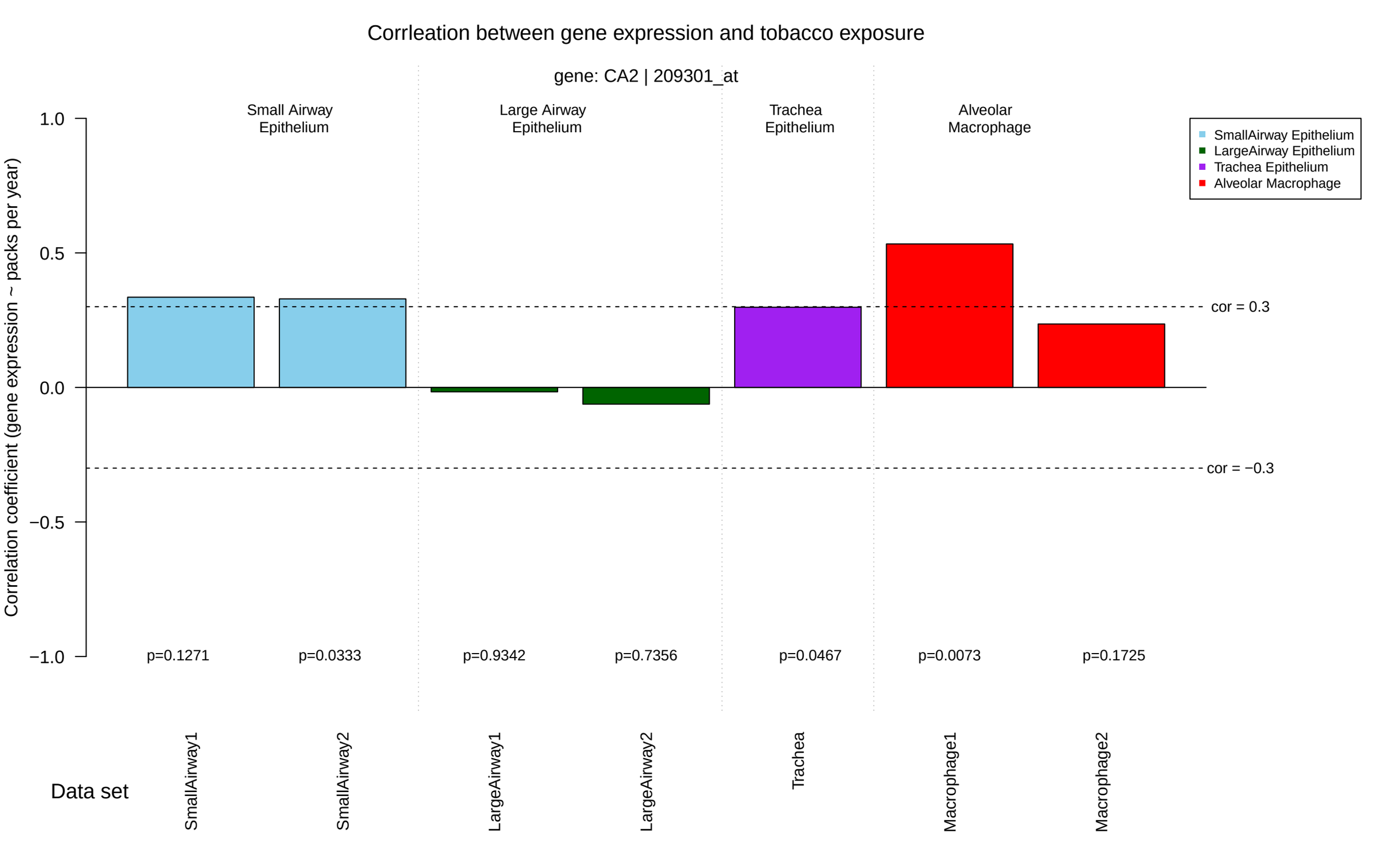

set: LargeAirway1 (p=0.2165)
CA2 | 209301_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7607)
CA2 | 209301_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1697)
CA2 | 209301_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0064)
CA2 | 209301_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0901)
CA2 | 209301_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA2 | 209301_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1271
p=0.0333
p=0.9342
p=0.7356
p=0.0467
p=0.0073
p=0.1725
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2